

VECTOR ANALYSIS

Scalar:

The quantity having only magnitude is called scalar. For example speed, distance etc.

Vector:

The quantity having both magnitude and direction is called a vector. For example velocity, displacement etc.

Following table shows the information of some vector ideas.

Scalar	$k, \phi (x, y, z), \lambda$
Vector	$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$
Magnitude	$ \vec{a} = \sqrt{a_1^2 + a_2^2 + a_3^2}$
Addition	$\vec{a} + \vec{b} = (a_1\hat{i} + b_1\hat{j}) + (a_1 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$
Subtraction	$\vec{a} - \vec{b} = (a_1\hat{i} - b_1\hat{j}) + (a_1 - b_2)\hat{j} + (a_3 - b_3)\hat{k}$
Scalar Multiplication	$\lambda \vec{a} = (\lambda a_1)\hat{i} + (\lambda a_2)\hat{j} + (\lambda a_3)\hat{k}$
Scalar Product	$\vec{a} \cdot \vec{b} = (a_1b_1)\hat{i} + (a_1b_2)\hat{j} + (a_3b_3)\hat{k}$ $= \vec{a} \vec{b} \cos \theta, \text{ where } \theta \text{ is the angle between } \vec{a} \text{ \& } \vec{b}.$
Vector Product	$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$ $= \vec{a} \vec{b} \sin \theta \hat{n}, \text{ where } \theta \text{ is the angle between } \vec{a} \text{ \& } \vec{b}. \text{ And } \hat{n} \text{ is the unit vector normal to the plane which contain } \vec{a} \text{ \& } \vec{b}.$
Directional derivative of ϕ in the direction of $\vec{a}$	$\frac{\vec{\nabla} \phi \cdot \vec{a}}{ \vec{a} }$
Scalar Triple product	$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$
Vector Triple product	$\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{c} \times \vec{a}) \cdot \vec{b} - (\vec{a} \times \vec{b}) \cdot \vec{c}$

Unit vector in the direction of vector $\vec{a}$	$\hat{a} = \frac{\vec{a}}{ \vec{a} }$
Direction angle of a vector $\vec{a}$	$\alpha = \cos^{-1} \frac{a_1}{ \vec{a} }$ (with x-axis) $\beta = \cos^{-1} \frac{a_2}{ \vec{a} }$ (with y-axis) $\beta = \cos^{-1} \frac{a_2}{ \vec{a} }$ (with z-axis)
Angle between two vectors $\vec{a}$ and $\vec{b}$	$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{ \vec{a} \vec{b} } \right)$
Projection of $\vec{a}$ along $\vec{b}$	$\frac{\vec{a} \cdot \vec{b}}{ \vec{b} }$
Vector differential operator del (nabla)	$\vec{\nabla} = \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k}$
Gradient	$\vec{\nabla} \phi = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k}$
Divergence	$\vec{\nabla} \cdot \vec{a} = \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}$
Curl	$\vec{\nabla} \times \vec{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$
Laplacian Operator	$\vec{\nabla}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

Rules of Vector Differentiation:

Let $\vec{f}(t)$, $\vec{g}(t)$ be vector functions of variable t and $u(t)$ be a scalar function of t . And let $\vec{a}$ be a constant vector.

$$1. \quad \frac{d\vec{a}}{dt} = 0$$

$$2. \quad \frac{d}{dt}(\vec{f} + \vec{g}) = \frac{d\vec{f}}{dt} + \frac{d\vec{g}}{dt}$$

$$3. \quad \frac{d}{dt}(u \vec{f}) = u \frac{d\vec{f}}{dt} + \frac{du}{dt} \vec{f}$$

$$4. \quad \frac{d}{dt}(\vec{f} \cdot \vec{g}) = \vec{f} \cdot \frac{d\vec{g}}{dt} + \frac{d\vec{f}}{dt} \cdot \vec{g}$$

$$5. \quad \frac{d}{dt}(\vec{f} \times \vec{g}) = \vec{f} \times \frac{d\vec{g}}{dt} + \frac{d\vec{f}}{dt} \times \vec{g}$$

$$6. \quad \frac{d}{dt}\left(\frac{\vec{f}}{u}\right) = \frac{u \frac{d\vec{f}}{dt} - \frac{du}{dt} \vec{f}}{u^2}$$

$$7. \quad \frac{d}{dt}(\vec{f} \cdot \vec{f}' \times \vec{f}'') = \vec{f} \cdot \vec{f}'' \times \vec{f}'''$$

$$8. \quad \frac{d}{dt}(\vec{f} \cdot \vec{g} \times \vec{h}) = \vec{f} \cdot \vec{g} \times \frac{d\vec{h}}{dt} + \vec{f} \cdot \frac{d\vec{g}}{dt} \times \vec{h} + \frac{d\vec{f}}{dt} \cdot \vec{g} \times \vec{h}$$

$$9. \quad \frac{d}{dt}(\vec{f} \times (\vec{g} \times \vec{h})) = \vec{f} \times \left(\vec{g} \times \frac{d\vec{h}}{dt}\right) + \vec{f} \times \left(\frac{d\vec{g}}{dt} \times \vec{h}\right) + \frac{d\vec{f}}{dt} \times (\vec{g} \times \vec{h})$$

Rules of Vector Integration:

Let $\vec{f}(t)$, $\vec{g}(t)$ be vector functions of variable t and $\vec{a}$ be a constant vector and k be a scalar constant.

$$1. \quad \int [\vec{f}(t) + \vec{g}(t)] dt = \int \vec{f}(t) dt + \int \vec{g}(t) dt$$

$$2. \quad \int k \vec{f}(t) dt = k \int \vec{f}(t) dt$$

$$3. \quad \int \vec{a} \cdot \vec{f}(t) dt = \vec{a} \cdot \int \vec{f}(t) dt$$

$$4. \quad \int \vec{a} \times \vec{f}(t) dt = \vec{a} \times \int \vec{f}(t) dt$$

$$5. \quad \int \vec{f}(t) \cdot \vec{g}(t) + \vec{f}(t) \cdot \vec{g}(t) dt = [\vec{f}(t) \cdot \vec{g}(t)] + k$$

Formulas Involving $\vec{\nabla}$:

If $\vec{f}$ and $\vec{g}$ are differentiable vector functions, and k and m are differentiable scalar functions of position (x, y, z) , then

$$1. \quad \vec{\nabla}(k + m) = \vec{\nabla}k + \vec{\nabla}m$$

$$2. \quad \vec{\nabla} \cdot (\vec{f} + \vec{g}) = \vec{\nabla} \cdot \vec{f} + \vec{\nabla} \cdot \vec{g}$$

$$3. \quad \vec{\nabla} \times (\vec{f} + \vec{g}) = \vec{\nabla} \times \vec{f} + \vec{\nabla} \times \vec{g}$$

4. $\vec{\nabla} (k \vec{f}) = (\vec{\nabla} k) \cdot \vec{f} + k (\vec{\nabla} \cdot \vec{f})$
5. $\vec{\nabla} \times (k \vec{f}) = (\vec{\nabla} k) \times \vec{f} + k (\vec{\nabla} \times \vec{f})$
6. $\vec{\nabla} \cdot (\vec{f} \times \vec{g}) = \vec{g} \cdot (\vec{\nabla} \times \vec{f}) - \vec{f} \cdot (\vec{\nabla} \times \vec{g})$
7. $\vec{\nabla} \times (\vec{f} \times \vec{g}) = (\vec{g} \cdot \vec{\nabla}) \vec{f} - \vec{g} (\vec{\nabla} \cdot \vec{f}) - (\vec{f} \cdot \vec{\nabla}) \vec{g} + \vec{f} (\vec{\nabla} \cdot \vec{g})$
8. $\vec{\nabla} (\vec{f} \cdot \vec{g}) = (\vec{g} \cdot \vec{\nabla}) \vec{f} + (\vec{f} \cdot \vec{\nabla}) \vec{g} + \vec{g} \times (\vec{\nabla} \times \vec{f}) + \vec{f} \times (\vec{\nabla} \times \vec{g})$
9. $\vec{\nabla} \cdot (\vec{\nabla} k) = \nabla^2 k$
10. $\vec{\nabla} \times (\vec{\nabla} k) = 0$
11. $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{f}) = 0$
12. $\vec{\nabla} \times (\vec{\nabla} \times \vec{f}) = (\vec{\nabla} \cdot \vec{f}) - \nabla^2 \vec{f}$

Divergence Theorem of Gauss:

If V is the volume bounded by closed surface S and $\vec{f}$ is a vector function of position with continuous derivatives, then

$$\iiint_V \vec{\nabla} \cdot \vec{f} \, dv = \iint_S \vec{f} \cdot \vec{n} \, dS$$

Where $\vec{n}$ is the positive (outward drawn) normal to S .

Stokes Theorem:

If S is an open, two sided surface bounded by a closed, non-intersecting curve C (simple closed curve) then if $\vec{f}$ has continuous derivatives

$$\int_C \vec{f} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{f}) \cdot \vec{n} \, dS$$

Where C is traversed in the positive direction. The direction of C and $\vec{n}$ is the positive (outward drawn) normal to S .

Green's Theorem in the plane:

If R is a closed region of xy plane bounded by a simple closed curve C and if $\vec{f}$ and $\vec{g}$ are continuous functions of x and y having continuous derivatives in R , then

$$\int_C \vec{f} \, dx + \vec{g} \, dy = \iint_R \left(\frac{\partial \vec{g}}{\partial x} - \frac{\partial \vec{f}}{\partial y} \right) dx dy$$

Where C is traversed in the positive (counter clockwise) direction.

Multiple Choice Questions:

1. Which of the following is not a vector quantity.
(a) velocity (b) displacement (c) speed (d) acceleration
2. For two vectors $\vec{a}$ & $\vec{b}$ which of the following is true.
(a) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (b) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
(c) $|\vec{a} - \vec{b}| = |\vec{b} - \vec{a}|$ (d) All of these
3. Which of the following vector has magnitude one.
(a) Null vector (b) Free vector (c) unit vector (d) position vector
4. If for two vectors $\vec{a}$ & $\vec{b}$, we have $\vec{a} = \lambda \vec{b}$, $\lambda \in \mathbb{R}$. Then $\vec{a}$ & $\vec{b}$ are
(a) collinear vector (b) along the same line
(c) parallel vectors (d) all of these
5. A vector whose position is not fixed in the space is
(a) Null vector (b) free vector (c) localised vector (d) None
6. A vector whose initial position is at origin is
(a) free vector (b) position vector
(c) unit vector (d) none
7. A vector of magnitude 5 in the discrete of $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$.
(a) $5\hat{a}$ (b) $-5\hat{a}$
(c) $\frac{5}{\sqrt{14}}\hat{i} + \frac{10}{\sqrt{14}}\hat{j} + \frac{15}{\sqrt{14}}\hat{k}$ (d) All of these
8. If α, β, γ are direction cosine of a vector, then which of the following is not true.
(a) $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ (b) $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$
(c) $\tan^2 \alpha + \tan^2 \beta + \tan^2 \gamma = 3$ (d) $\sin^2 \alpha - \cos^2 \beta + \sin^2 \gamma = 1$
9. The direction cosines of x-axis are
(a) 1, 0, 0 (b) 0, 1, 0 (c) 0, 0, 1 (d) 1, 0, 1
10. The direction cosines of y-axis are
(a) 1, 0, 0 (b) 0, 1, 0 (c) 0, 0, 1 (d) 1, 0, 1
11. The direction cosines of z-axis are
(a) 1, 0, 0 (b) 0, 1, 0 (c) 0, 0, 1 (d) 1, 0, 1
12. Which of the following can be a direction angle of a vector
(a) $30^\circ, 45^\circ, 60^\circ$ (b) $30^\circ, 45^\circ, 90^\circ$
(c) $45^\circ, 45^\circ, 60^\circ$ (d) $60^\circ, 45^\circ, 60^\circ$

13. If a vector makes an angle 45° with x-axis and 30° with y-axis. The its angle with z-axis will be
 (a) 105° (b) 90° (c) $\cos^{-1}\left(\frac{1}{\sqrt{6}}\right)$ (d) 60°
14. If a vector makes an angle of equal magnitude with there axis then its direction consines are
 (a) $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (b) $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$
 (c) $45^\circ, 60^\circ, 90^\circ$ (d) $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$
15. The scalar product of two vector is zero, then the vectors are
 (a) positive (b) parallel (c) collinear (d) perpendicular
16. Which of the following statement is true for vectors $\vec{a}$ & $\vec{b}$
 (a) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ (b) $\vec{a} \cdot (\vec{a} + \vec{b}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
 (c) $\vec{a} \cdot \vec{a} \geq 0$ (d) All of these
17. Which of the following is a vector quantity.
 (a) $\vec{a} \cdot \vec{b}$ (b) $\vec{a} \times \vec{b}$ (c) $|\vec{a} \times \vec{b}|$ (d) $\vec{a} \times \vec{a}$
18. Two vectors $\vec{a}$ & $\vec{b}$ are parallel if
 (a) $\vec{a} \cdot \vec{b} = 0$ (b) $\vec{a} \times \vec{b} = 0$ (c) $\vec{a} \cdot \vec{a} = 0$ (d) $\vec{b} \cdot \vec{b} = 0$
19. Let θ be an angle between vectors $\vec{a}$ & $\vec{b}$. Then
 (a) $\theta = \sin^{-1}\left(\frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}\right)$ (b) $\theta = \cos^{-1}\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}\right)$
 (c) both (a) & (b) (d) None
20. $|\vec{a} \times \vec{b}| =$
 (a) Area of triangle (b) Area of circle
 (c) Area of parallelogram (d) Area of rectangle
21. If two vectors $\vec{a}$ & $\vec{b}$ are perpendicular. Then which of the following is true.
 (a) The angle between $\vec{a}$ & $\vec{b}$ is 90° (b) $\vec{a} \cdot \vec{b} = 0$
 (c) $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$ (d) All of these
22. Let $\vec{a}$ and $\vec{b}$ two vectors such that $|\vec{a}| = |\vec{b}| = |\vec{a} - \vec{b}|$. Then angle between $\vec{a}$ & $\vec{b}$ will be
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$
23. For any two vectors $\vec{a}$ & $\vec{b}$, which of the following is true.
 (a) $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$ (b) $|\vec{a}| - |\vec{b}| \leq |\vec{a} + \vec{b}|$
 (c) $|\vec{a}| - |\vec{b}| \leq |\vec{a}| + |\vec{b}|$ (d) All of these

24. Which of the following is a unit vector perpendicular $\vec{a}$ & $\vec{b}$
- (a) $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$ (b) $\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$ (c) $\vec{a} \times \vec{b}$ (d) $\hat{a} \cdot \hat{b}$
25. What is the area of parallelogram whose adjacent sides are $2\hat{i} - \hat{j} + 3\hat{k}$ and $3\hat{i} + 7\hat{j} - \hat{k}$.
- (a) $\sqrt{720}$ (b) $\sqrt{810}$ (c) $\sqrt{905}$ (d) $\sqrt{1024}$
26. $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) =$
- (a) $2\vec{a}$ (b) $2\vec{b}$ (c) $2(\vec{a} \times \vec{b})$ (d) $2\vec{a}\vec{b}$
27. If $\vec{a}$, $\vec{b}$, $\vec{c}$ represent the triangle $\triangle ABC$. Then $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$ is perpendicular to
- (a) $\vec{a} \times \vec{b} \times \vec{c}$ (b) $\vec{a} \cdot \vec{b} \times \vec{c}$ (c) $\triangle ABC$ (d) All of these
28. In the scalar triple product $\vec{a} \cdot \vec{b} \times \vec{c} =$
- (a) $-\vec{a} \cdot (\vec{c} \times \vec{b})$ (b) $\vec{b} \cdot \vec{c} \times \vec{a}$
 (c) $\vec{c} \cdot \vec{a} \times \vec{b}$ (d) All of these
29. For vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, the product $\vec{a} \cdot \vec{b} \times \vec{a} =$
- (a) $2\vec{a}$ (b) $\vec{b}$ (c) 0 (d) 1
30. The scalar triple product $\vec{a} \cdot \vec{b} \times \vec{c}$ represents
- (a) area of triangle (b) volume of parallelogram
 (c) volume of tetrahedron (d) volume of parallelepiped
31. For vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ which of the following is not correct
- (a) $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$ (b) $\vec{a} \times (\vec{b} \times \vec{c}) = -(\vec{b} \times \vec{c}) \times \vec{a}$
 (c) $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$ (d) $\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$
32. If $\vec{f}(t)$ and $\vec{g}(t)$ are vector function, then which of the following is correct.
- (a) $\frac{d}{dt}(\vec{f} + \vec{g}) = \frac{d\vec{f}}{dt} + \frac{d\vec{g}}{dt}$ (b) $\frac{d}{dt}(\vec{f} \cdot \vec{g}) = \vec{f} \frac{d\vec{g}}{dt} + \frac{d\vec{f}}{dt} \cdot \vec{g}$
 (c) $\frac{d}{dt}(\vec{f} \times \vec{g}) = \vec{f} \frac{d\vec{g}}{dt} + \frac{d\vec{f}}{dt} \times \vec{g}$ (d) All of these
33. If $\vec{f}(t)$ and $\vec{g}(t)$ are vector function, then which of the following is correct
- (a) $\int [\vec{f}(t) + \vec{g}(t)] dt = \int \vec{f}(t) dt + \int \vec{g}(t) dt$
 (b) $\int k\vec{f}(t) dt = k \int \vec{f}(t) dt$ where k is scalar
 (c) $\int \vec{r} \cdot \vec{f}(t) dt = \vec{r} \cdot \int \vec{f}(t) dt$ where, $\vec{r}$ is constant vector
 (d) All of these

34. If $\vec{f}(t) = a \cos t \underline{i} + a \sin t \underline{j} + bt \underline{k}$, then $\vec{f}' \cdot \vec{f}'' \times \vec{f}''' =$
 (a) 0 (b) 1 (c) a^2b (d) ab^2
35. If $\vec{f}(t)$ is a vector function, then $\frac{d}{dt}(\vec{f} \cdot \vec{f}' \times \vec{f}'') =$
 (a) 0 (b) $\vec{f} \cdot \vec{f}' \times \vec{f}''$ (c) $\vec{f}' \cdot \vec{f}'' \times \vec{f}'''$ (d) $\vec{f} \cdot \vec{f}' \times \vec{f}'''$
36. If $\vec{r}(t) = \underline{a} \cos nt + \underline{b} \sin nt$ then $\vec{r} \times \frac{d\vec{r}}{dt} =$
 (a) 0 (b) n (c) $n(\underline{a} \times \underline{b})$ (d) $n(\underline{a} \cdot \underline{b})$
37. Which of the following is the solution of vector differential equation $\frac{d^2\vec{V}}{dt^2} = \vec{a}$
 (a) $\vec{V} = \frac{at^2}{2} + \vec{c}$ (b) $\vec{V} = \frac{at^2}{2} + \vec{b}t + \vec{c}$
 (c) $\vec{V} = \frac{\vec{a}t^2}{2} + \frac{\vec{b}t}{2} + \vec{c}$ (d) None
38. If $\vec{r}(t) = 5t^2\underline{i} + t\underline{j} - t^3\underline{k}$, then $\int_1^2 \left(\vec{r} \times \frac{d^2\vec{r}}{dt^2} \right) dt =$
 (a) $-14\underline{i} + 75\underline{j} - 15\underline{k}$ (b) 0
 (c) 1 (d) 2
39. If $\vec{r}(2) = 2\underline{i} - \underline{j} + 2\underline{k}$ and $\vec{r}(3) = 4\underline{i} - 2\underline{j} + 3\underline{k}$ then $\int_2^3 \vec{r} \cdot \frac{d\vec{r}}{dt} dt =$
 (a) 0 (b) 5 (c) 10 (d) 6
40. If $\vec{r}(0) = 5\underline{i} - 3\underline{j} + 2\underline{k}$, $\vec{r}(7) = \underline{i} + 8\underline{j} + 9\underline{k}$ then $\int_0^7 \vec{r} \cdot \frac{d\vec{r}}{dt} dt =$
 (a) 0 (b) 7 (c) 28 (d) 54
41. The differential operator $\nabla =$
 (a) $\frac{\partial \vec{r}}{\partial x} + \frac{\partial \vec{r}}{\partial y} + \frac{\partial \vec{r}}{\partial z}$ (b) $\frac{\partial}{\partial x} \underline{i} + \frac{\partial}{\partial y} \underline{j} + \frac{\partial}{\partial z} \underline{k}$
 (c) $\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$ (d) None
42. Which of the following is correct
 (a) $\nabla(f+g) = \nabla f + \nabla g$ (b) $\nabla(f-g) = \nabla f - \nabla g$
 (c) $\nabla(fg) = f\nabla g + g\nabla f$ (d) All of these

43. If $\vec{r} = x\vec{i} + y\vec{j} + 3z\vec{k}$ then $\nabla r^n =$
 (a) $n r^{n-1} \vec{r}$ (b) $n r^{n-2} \vec{r}$
 (c) $n r^{n-3} \vec{r}$ (d) $n r^{n-4} \vec{r}$
44. Which of the following is a normal vector to surface ϕ
 (a) $\nabla \phi$ (b) $\nabla \cdot \phi$ (c) $\nabla \times \phi$ (d) None
45. Which of the following is the divergence of vector function $\vec{f}$.
 (a) $\nabla \vec{f}$ (b) $\nabla \cdot \vec{f}$ (c) $\nabla \times \vec{f}$ (d) none
46. If $\vec{a} + \vec{b} + \vec{c} = 0$. Then which of the following is true.
 (a) $\vec{a} \times \vec{b} = \vec{c}$ (b) $\vec{a} \times \vec{b} = \vec{c} \times \vec{a}$
 (c) $\vec{a} \times \vec{b} \times \vec{c} = 0$ (d) $\vec{a} \vec{b} \vec{c} = 0$
47. Which of the following is curl of a vector function $\vec{f}$.
 (a) $\nabla \vec{f}$ (b) $\nabla \cdot \vec{f}$ (c) $\nabla \times \vec{f}$ (d) none
48. Which of the following is equal to zero.
 (a) $\text{curl curl } \vec{A}$ (b) $\text{div curl } \vec{A}$
 (c) $\text{curl curl curl } \vec{A}$ (d) $\text{gradient } \vec{A}$
49. Which of the following is non-zero.
 (a) $\text{curl } r^n \vec{r}$ (b) $\text{cur} \left(\frac{\vec{r}}{r^2} \right)$
 (c) $\nabla \cdot \nabla \phi$ (d) $\nabla^2 \left(\frac{1}{r} \right)$
50. $\text{Curl} (\phi \text{ grad } \psi) + \text{curl} (\psi \text{ grad } \phi) =$
 (a) 1 (b) -1 (c) 2 (d) 0

Key To MCQ of Vector Analysis

1	c	2	d	3	c	4	d	5	b	6	b
7	c	8	c	9	a	10	b	11	c	12	d
13	c	14	b	15	d	16	d	17	b	18	b
19	b	20	c	21	d	22	b	23	d	24	b
25	b	26	c	27	c	28	d	29	c	30	d
31	d	32	d	33	d	34	c	35	d	36	b
37	b	38	a	39	c	40	d	41	b	42	d
43	b	44	a	45	b	46	b	47	c	48	b
49	c	50	d								